Team Details

- ➔ Team name – LONE CODER
- ➔ Team leader name – P HARI RAM NIKIL
- ➔ Problem Statement – AI IN CONSUMER EXPERIENCES

Brief about the idea

The Concept:

"Sentinel" is a next-generation consumer experience platform designed to solve the conflict between Personalization and Privacy. It is an on-device AI agent that acts as a intelligent personal shopper and lifestyle curator. Unlike current AI assistants that process data in the cloud, Sentinel operates **100% locally** on the user's machine.

How it Works:

The system uses a "Local Knowledge Graph" to analyze a user's private digital footprint such as local photo galleries (for style/fashion preferences), past digital receipts (for budget and brand loyalty), and calendars (for upcoming needs). By running a quantized Large Language Model (LLM) and Computer Vision models locally, it provides real-time, context-aware suggestions for products and services without a single byte of personal data ever leaving the device.

The Core Focus:

The idea shifts the consumer experience from **"Search & Discovery"** (where the user does the work) to **"Predictive Curation"** (where the AI does the work). For example, if a user has a wedding on their local calendar, Sentinel proactively suggests outfits from sustainable brands that match the user's existing wardrobe all while the user is offline or concerned about data security.

Opportunities

- ➔ How different is it from any of the other existing ideas?
- ➔ How will it be able to solve the problem?
- ➔ USP of the proposed solution

How different is it?
Unlike cloud-based AIs (ChatGPT/Siri) that store personal data on external servers, this solution operates entirely within the user's hardware. It utilizes the AMD Ryzen AI NPU to process private information locally, ensuring zero data leakage.

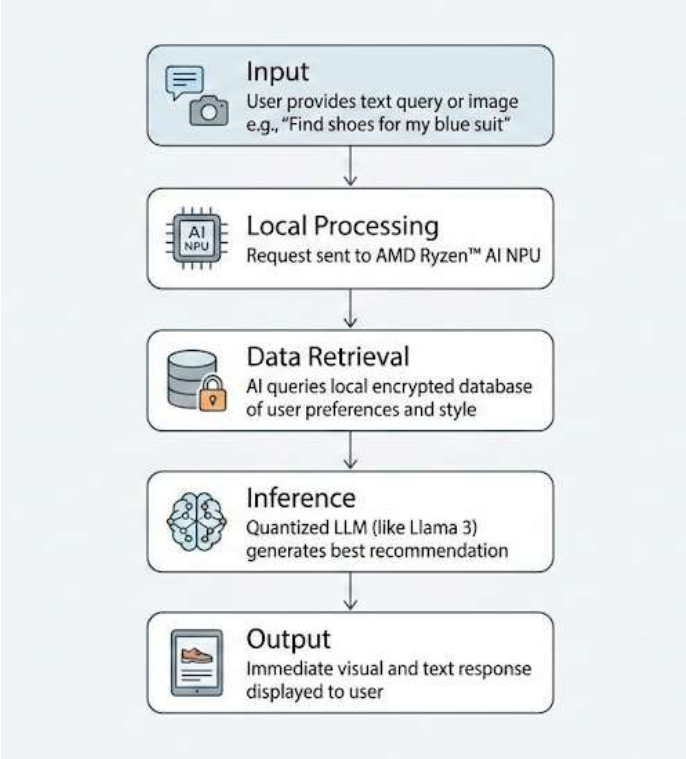
How will it solve the problem? It bridges the "Privacy Performance Gap" by removing the latency of cloud communication and the risk of data breaches. It provides instant, context aware suggestions while the device remains offline.

USP of the proposed solution: "Zero Cloud Intelligence." It is a high speed, secure, and energy efficient consumer experience.

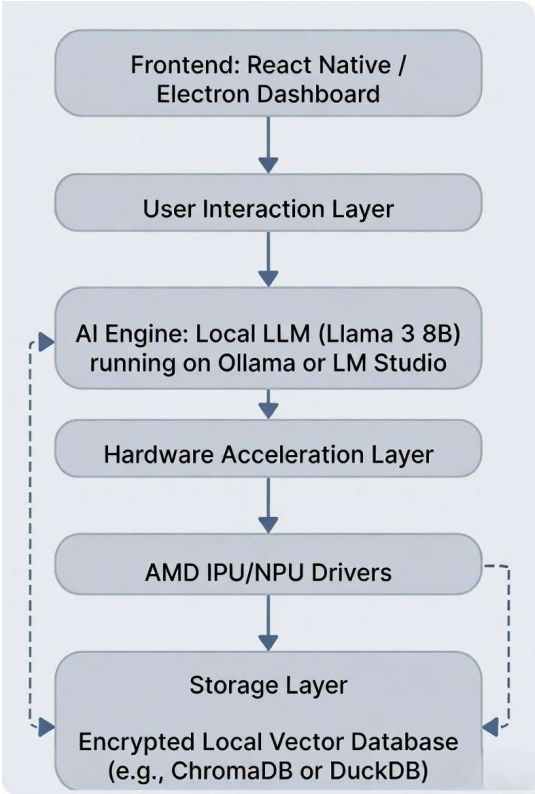
List of features offered by the solution

- **Local Visual Stylist:** Uses computer vision to match new products with a user's locally stored "Digital Wardrobe".
- **Privacy Safe Budgeting:** Scans local digital receipts to offer financial advice without requiring bank logins.
- **Contextual Shopping Reminders:** Syncs with local calendars to suggest gifts or outfits for upcoming events.
- **Offline Mode:** Fully functional without an internet connection, ensuring reliability anywhere.

Process flow diagram or Use-case diagram



Architecture diagram of the proposed solution



Technologies to be used in the solution

- **Large Language Model (LLM):** We will utilize Llama 3 (8B) or Mistral-v0.3, specifically 4-bit quantized versions (GGUF/EXL2 formats). This reduces the memory footprint, allowing the model to fit entirely within the dedicated NPU/GPU memory for lightning-fast inference without lagging the OS.

- **Runtime & Acceleration:** AMD ROCm Used to provide a Linux-based computational foundation that allows our vision models to communicate directly with the GPU hardware.

- **Ryzen™ AI Software:** We will use the ONNX Runtime AI execution provider to partition the model. This ensures the NPU handles the constant "background" AI tasks while keeping the CPU free for UI responsiveness.

- **Full-Stack Development:**
  Backend: Python with FastAPI for high performance asynchronous processing of local data.

  Frontend: React.js or Electron to build a sleek, desktop integrated dashboard that feels native to Windows/Linux.

Usage of AMD Products/Solutions

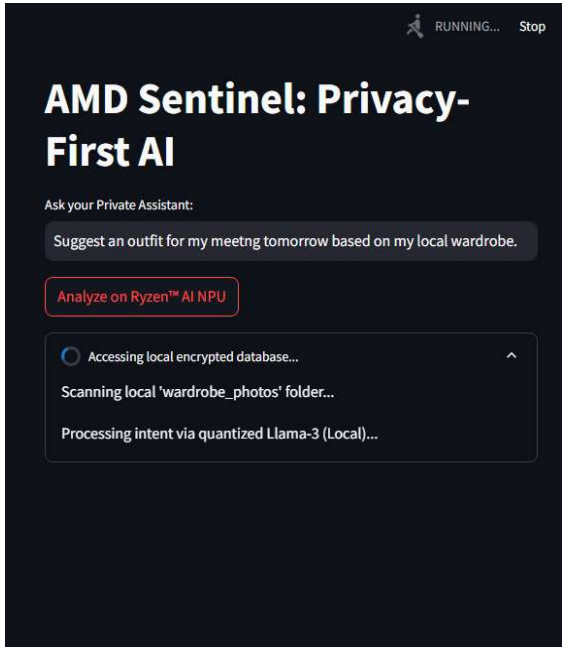
- **AMD Ryzen™ AI (NPU):** This is the heart of the project. By offloading the LLM and Embedding models to the NPU, we achieve "Always On" intelligence. This reduces CPU/GPU power draw by up to 40%, preventing laptop fans from spinning up and preserving battery life during long shopping sessions.
- **AMD Radeon™ Graphics:** For the "Consumer Experience" side, we utilize Radeon's RDNA™ architecture to handle high-fidelity 3D garment or furniture previews. This allows the consumer to see 3D rendered products with realistic lighting (Ray Tracing) directly in the app.
- **AMD XDNA™ Architecture:** We leverage XDNA's spatial partitioning to run Multimodal AI. This means the system can process an image (Computer Vision) and a user's text query (NLP) simultaneously on the NPU without a bottleneck, providing a fluid, lag head free experience.

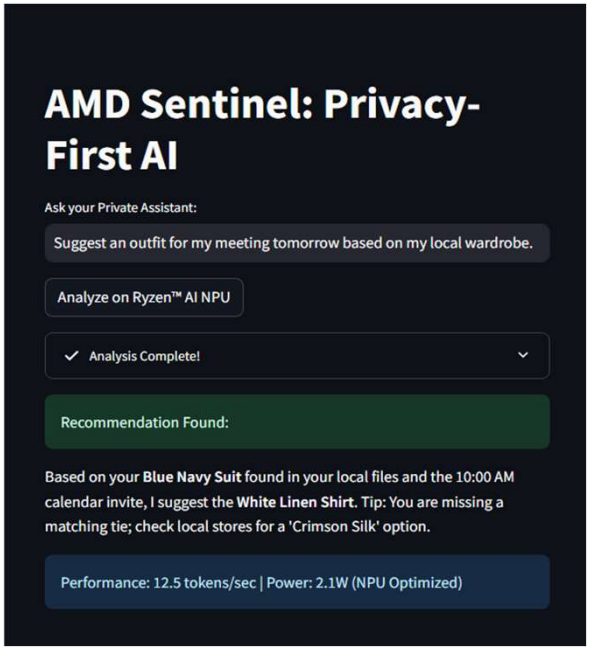
Estimated implementation cost (optional)

- **Development Cost** By leveraging the open source ecosystem specifically Hugging Face for models and AMD's Vitis there are zero licensing fees for the core intelligence of the app.
- **Hardware Efficiency:** The solution is designed to run on Consumer Off The Shelf hardware. There is no need for expensive, enterprise grade A100 GPUs. Any laptop featuring an AMD Ryzen 7040/8040 series or newer is a ready to use AI Supercomputer.
- **Scalability & Maintenance:** Since the AI is On Device, there are Zero Server Costs ($0/month). For the developer (Lone Coder) and the retailer, this eliminates the massive "Compute Bill" usually associated with scaling AI to millions of users.
- **Total Economic Impact:** A high margin, low overhead solution that turns a standard AMD laptop into a premium, private personal concierge.

Prototype Assets (Optional)

- ➔ GitHub Public Repository Link : https://github.com/nikilearth/AMD-SlingShot-Sentinel
- ➔ Demo Video Link (Max: 3 Minutes)





AMD Slingshot

HUMAN *IMAGINATION*
BUILT WITH *AI*

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Thank you!